

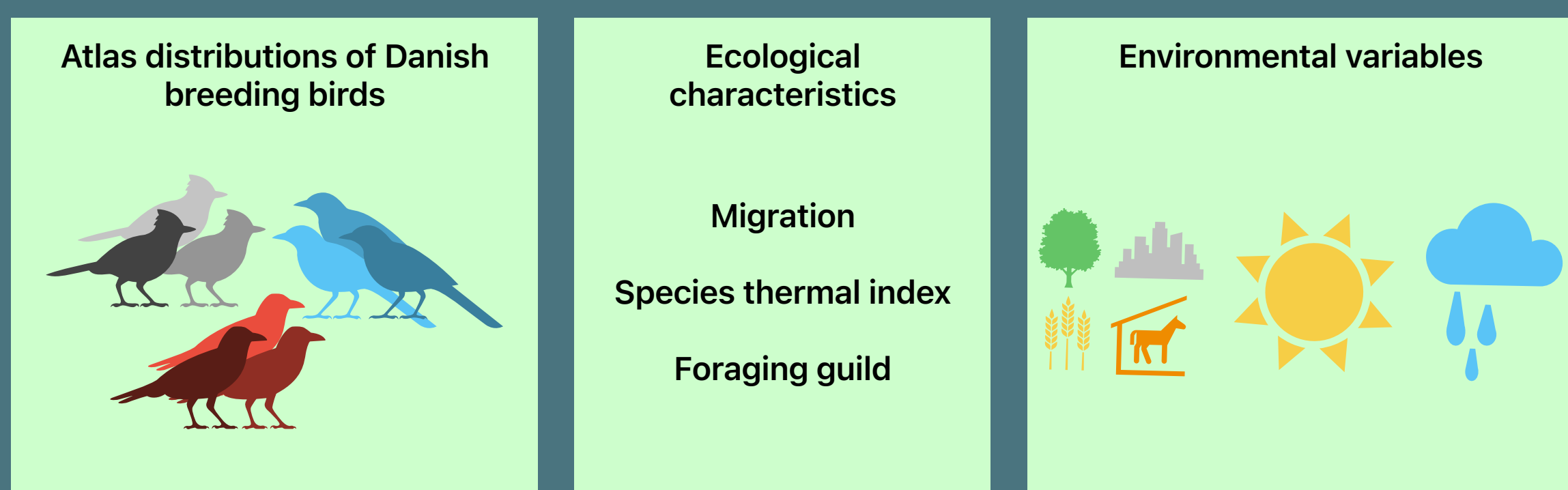
DATA

METHOD

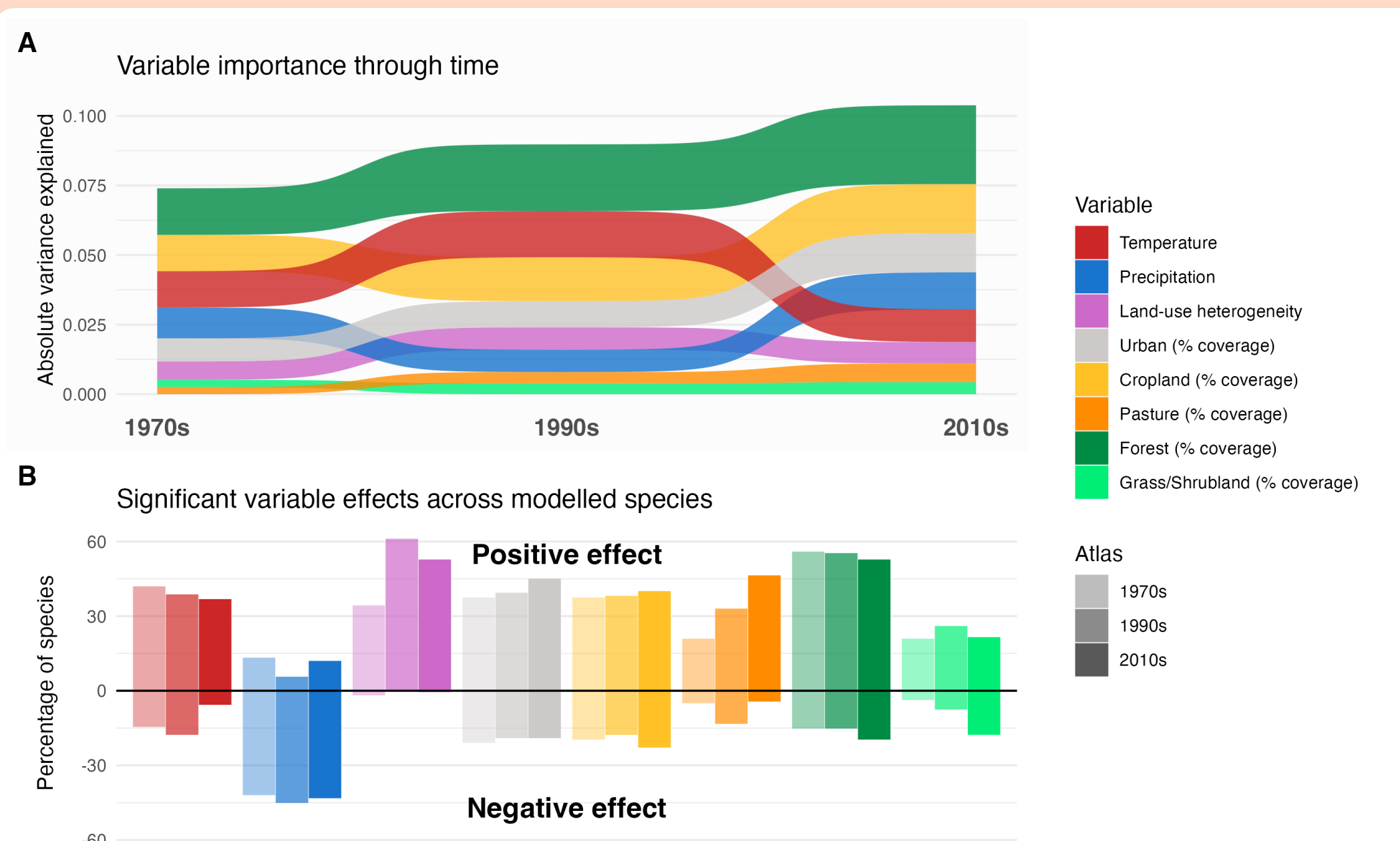
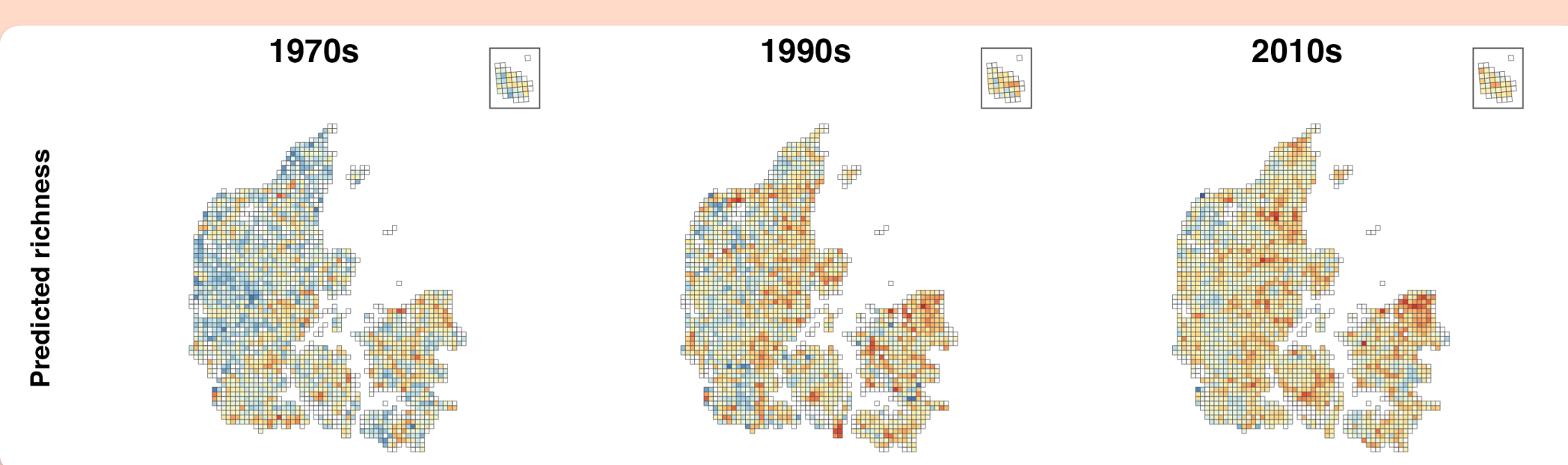
MODEL OUTPUT

1

Influence of environment on Danish avian communities



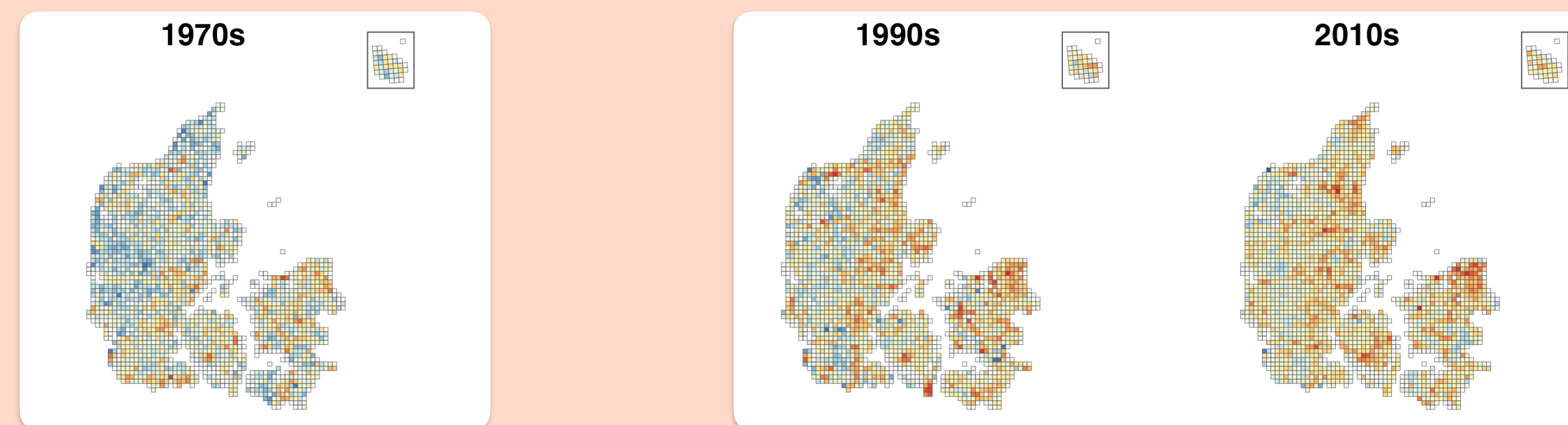
Species distributions and influence of environment



2

Can we cluster distinct clusters of communities? Are the clusters consistent through time?

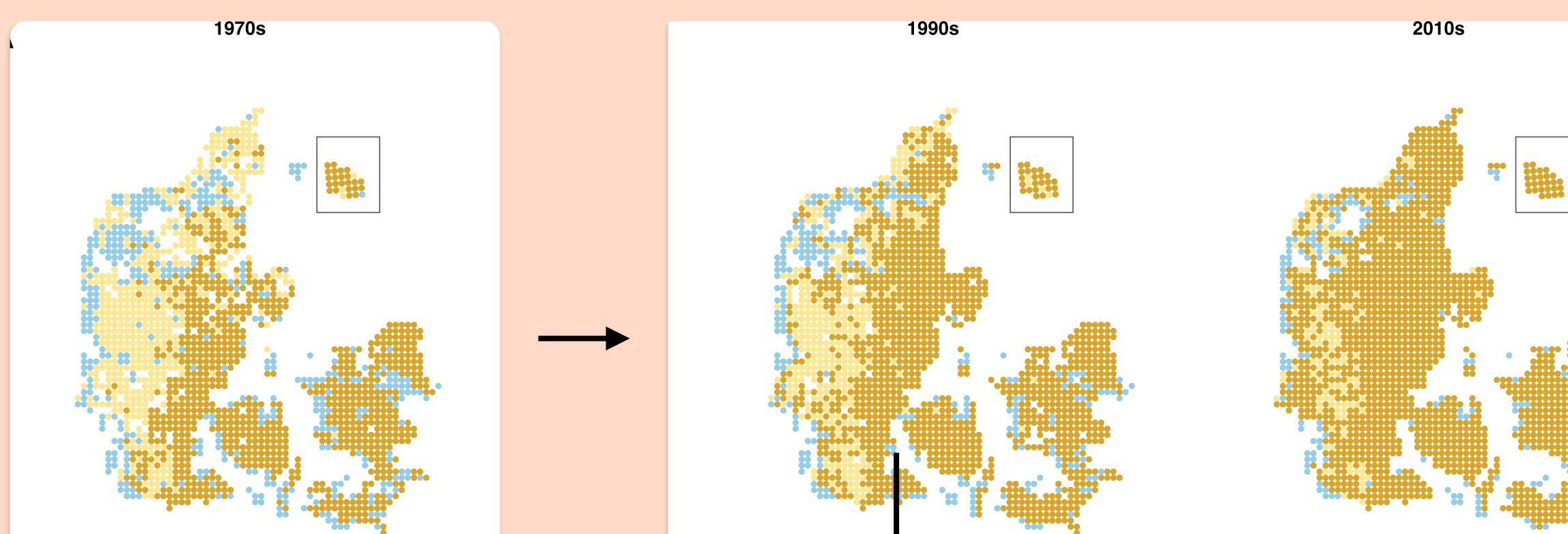
JSDM predictions



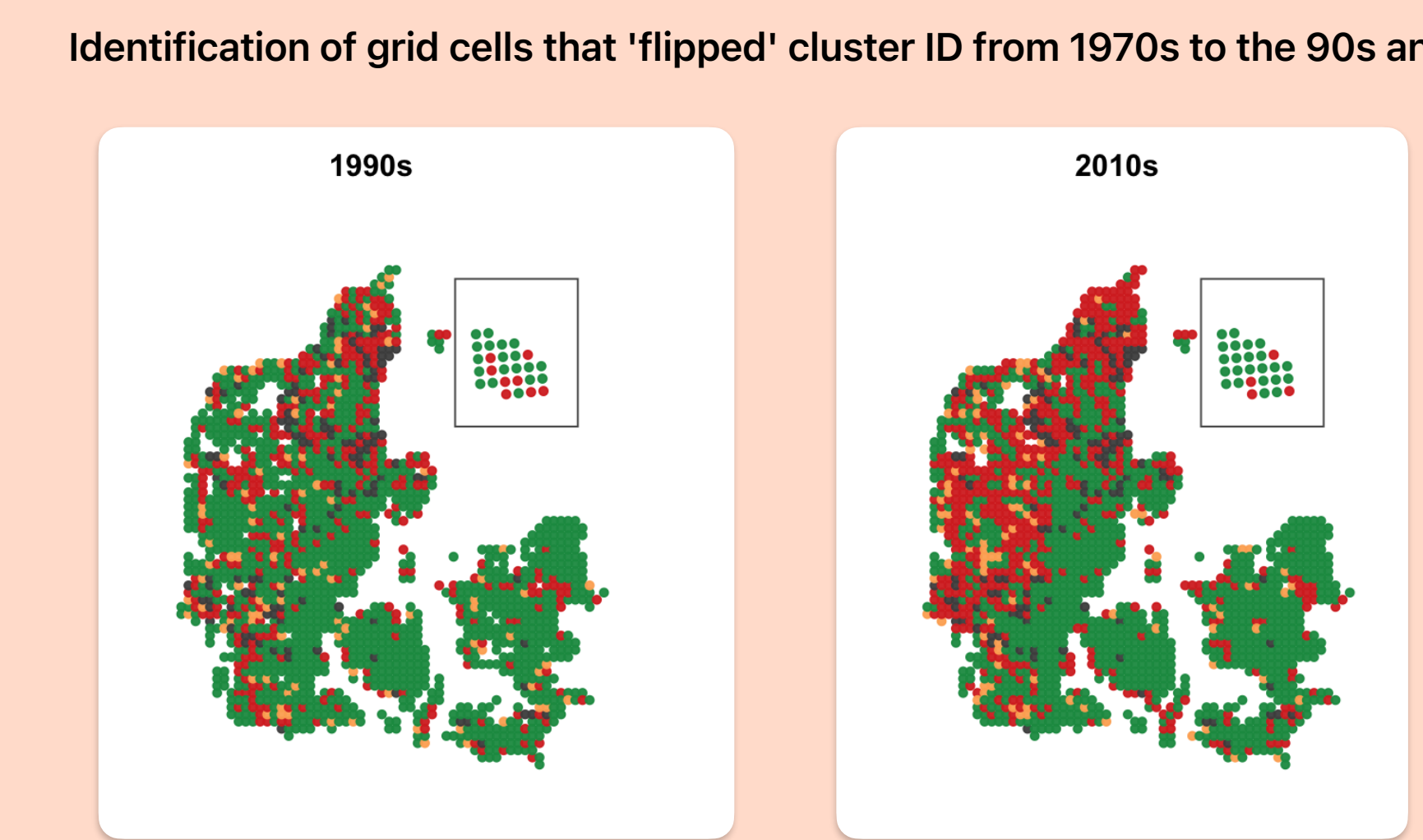
1. K-means clustering for regions of similar community composition

2. With 1970s clusters as a reference, which cluster are the grid cells in 1990 and 2010 most similar to?

Identification of grid cells with similar avian communities



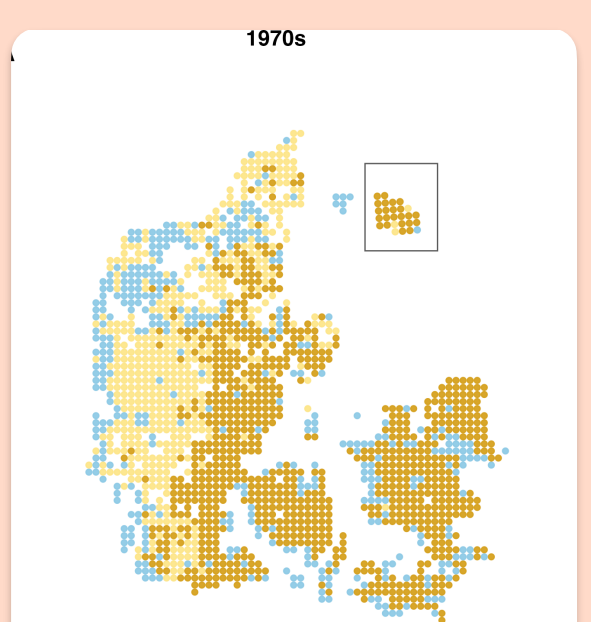
Comparison of cluster ID for each grid cell at different moments in time



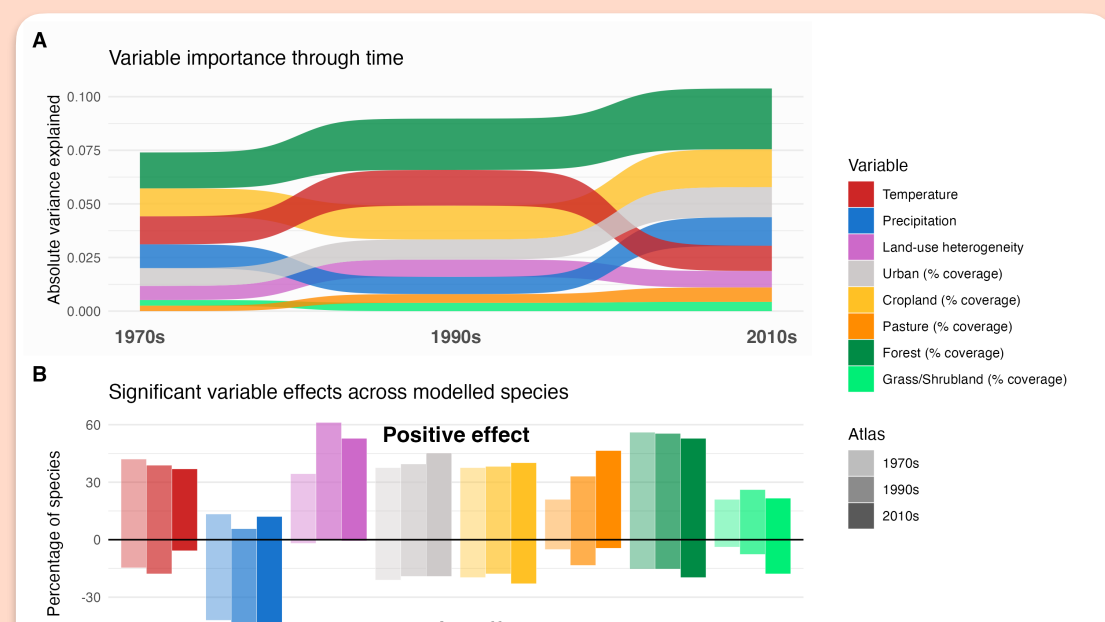
2.5

Environmental fingerprint of different clusters

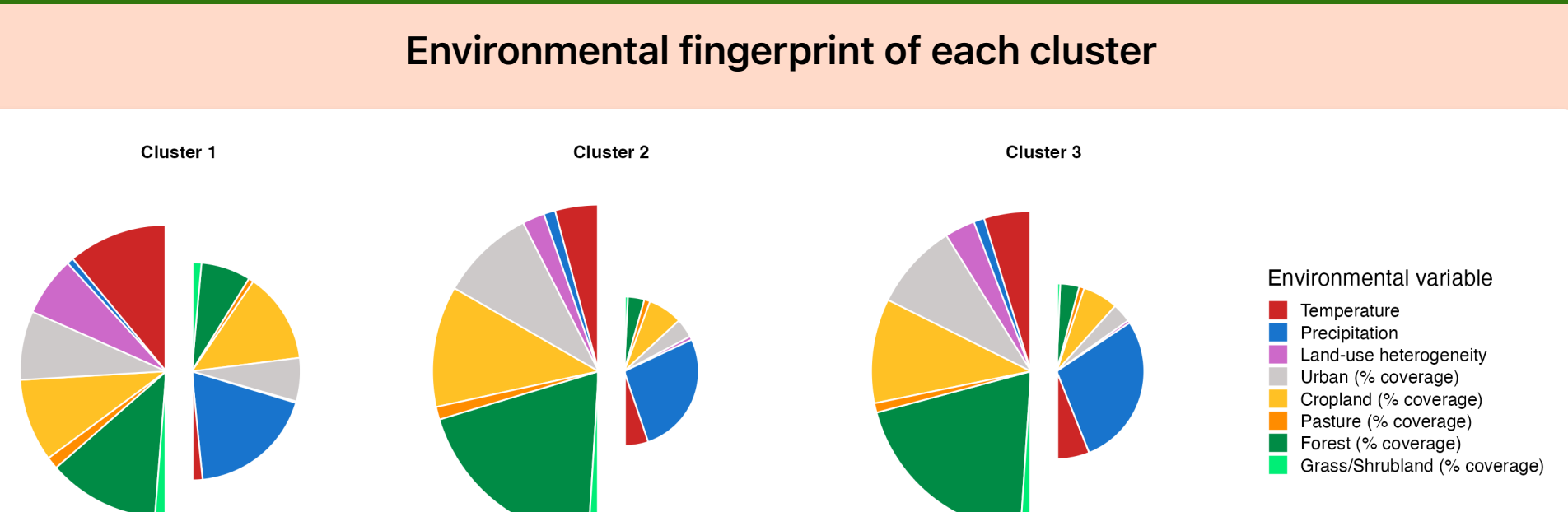
1970s community clusters



JSDM environmental influence



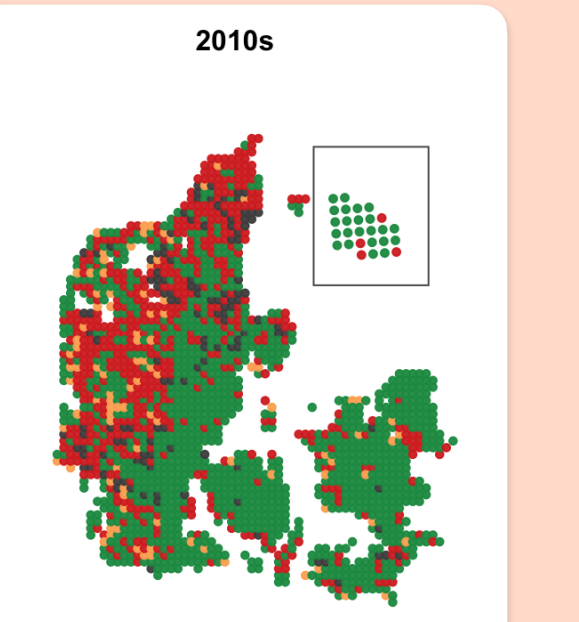
Cluster-level environmental fingerprint - species environmental drivers weighed by species probability in said cluster



3

> 35% of clusters 'flip' into the dominant cluster. What is underpinning 'flipping' from one cluster ID in 1970s to the dominant cluster in 2010s?

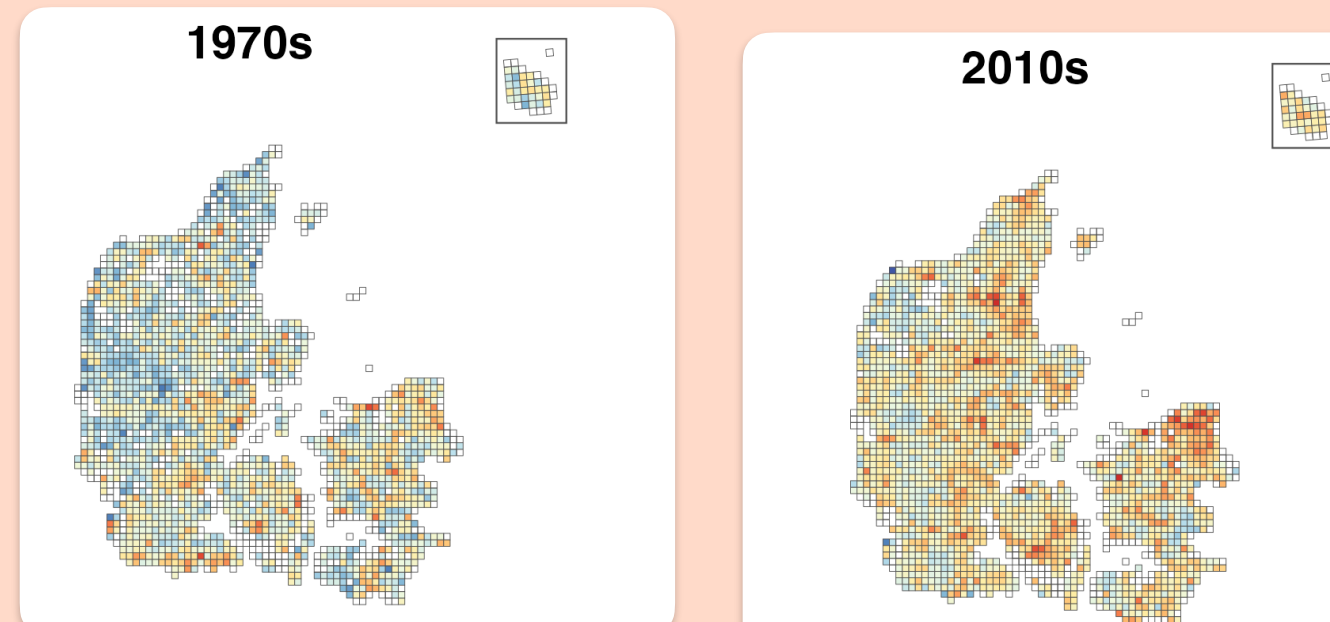
Grid cells that 'flipped' into dominant profile from 1970s



Ecological characteristics

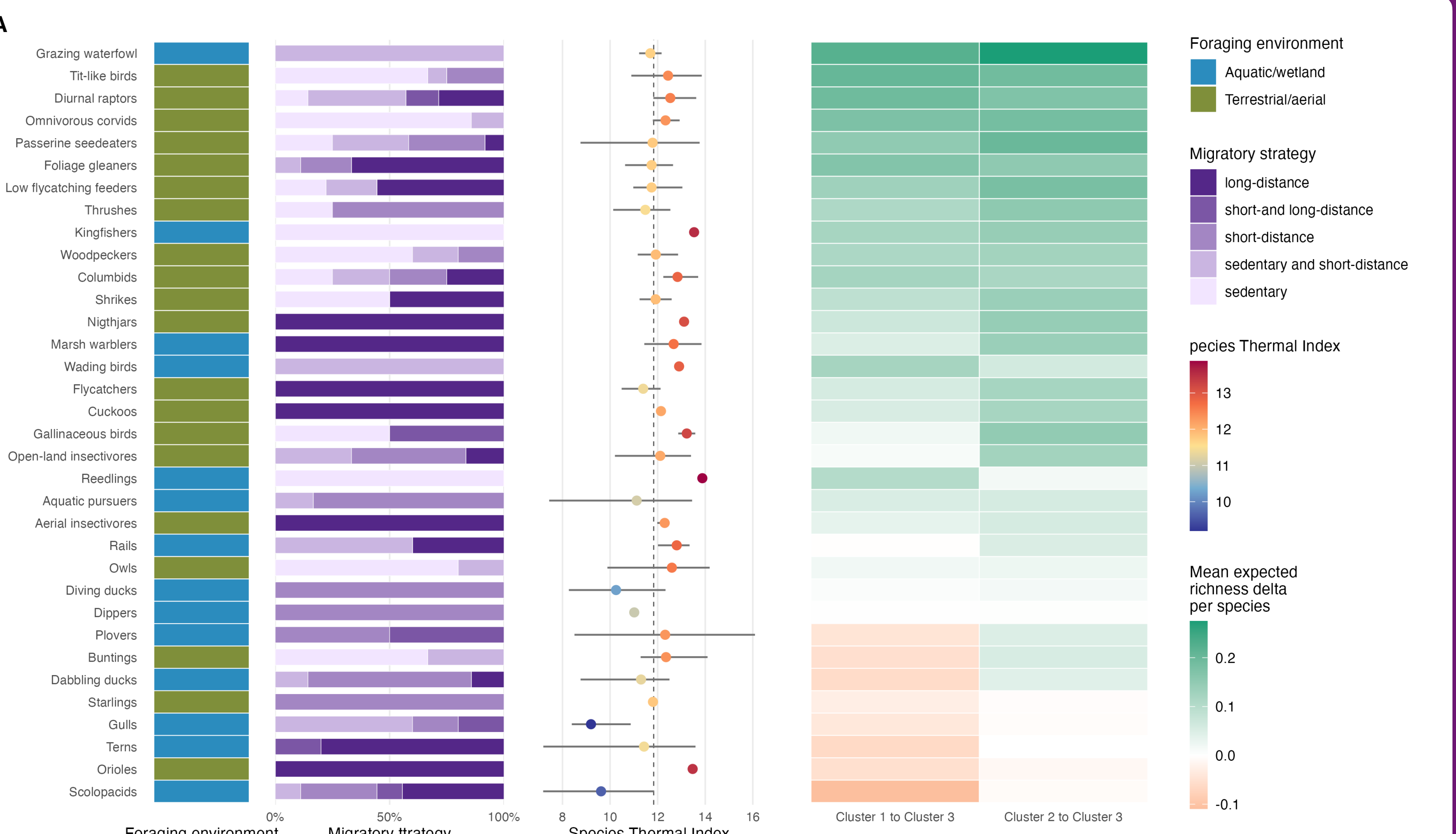
Migration
Species thermal index
Foraging guild

Grid-level, species level changes in probability of occurrence 1970s to 2010s



Are certain groups of species more inclined to increase / decrease in grid cells that flipped from cluster 1 or 2 to cluster 3 (the 'dominant' cluster), from the 1970s to the 2010s?

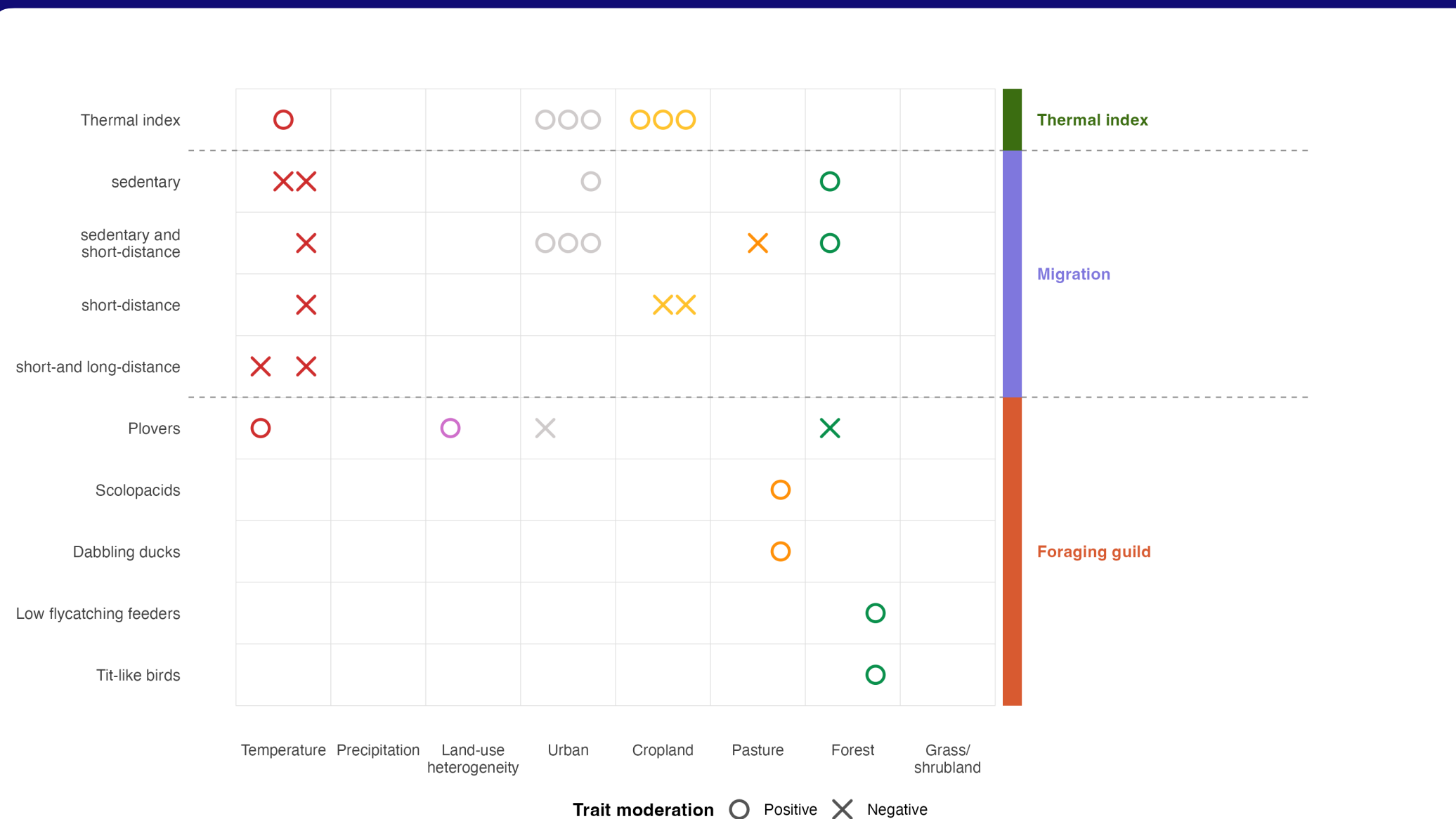
- Across grid cells that flipped to the dominant cluster, per species, get the average gain/loss in predicted probability of occurrence
- Get the gains/losses of all species in each 'trait group'
- normalize to an 'average gain/loss of this trait group per species' so the results aren't skewed by different group sizes



3.5

Can any changes of specific 'groups' of species gains/loss be explained through the lens of environmental variables

Joint SDMs



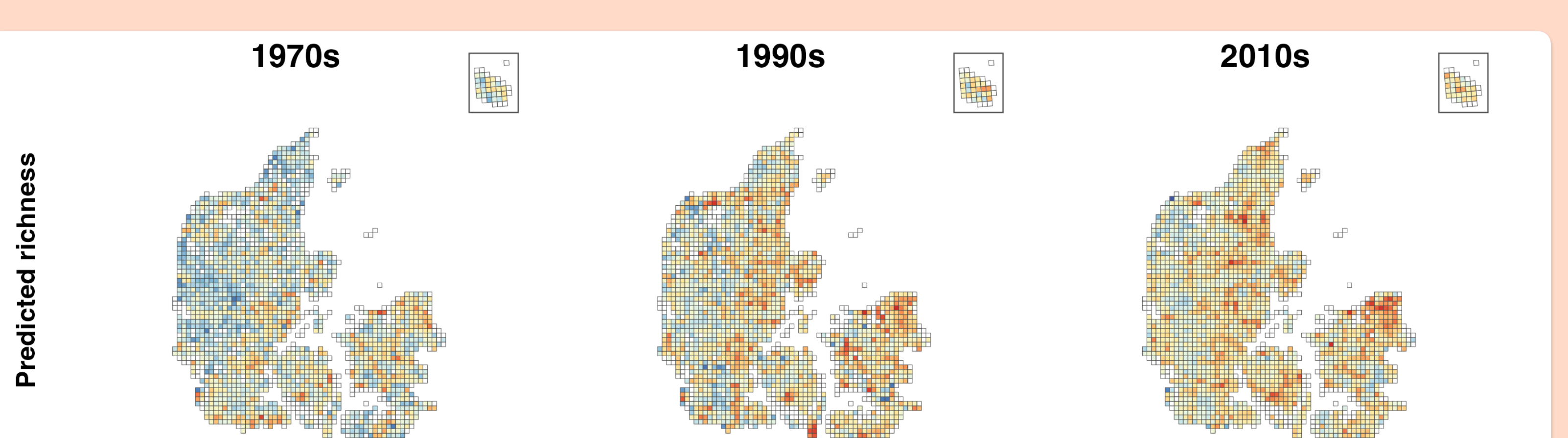
4

What is the spatial fingerprint of warm-affinity bird expansion? Is community warming driven purely by warm-affinity expansions or also by cold-affinity losses?

Ecological characteristics

Species thermal index

Predicted species distributions



Community weighted mean of species thermal index

Identification of grid cells dominated by cold vs warm affinity species

